5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T\to\infty}\text{var}(b_2|\mathbf{x}) = 0$.

a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \ldots, T$. Note that $\mathbf{x} = (1, 2, \ldots, T)$, $\sum_{t=1}^{T}(t-\bar{t})^2 = \sum_{t=1}^{T} t^2 - \left(\sum_{t=1}^{T} t\right)^2 / T$, $\sum_{t=1}^{T} t = T(T+1)/2$ and $\sum_{t=1}^{T} t^2 = T(T+1)(2T+1)/6$.

b. $y_t = \beta_1 + \beta_2(0.5)^t + e_t, t = 1, 2, \ldots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \ldots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \cdots + r^n = \frac{r(1-r^n)}{1-r}$$

$x_t = 0.5^t$
r 的等比数列

c. Provide an intuitive explanation for these results.

5.5

a.

$$\text{var}(b_2|x) = \frac{\sigma^2}{\sum_{t=1}^{T}(t-\bar{t})^2} = \frac{\sigma^2}{\sum_{t=1}^{T} t^2 - (\sum_{t=1}^{T} t)^2/T} = \frac{\sigma^2}{\frac{T(T+1)(2T+1)}{6} - \frac{[\frac{T(T+1)}{2}]^2}{T}}$$

$$= \frac{\sigma^2}{\frac{1}{6}(2T^3+3T^2+T) - \frac{1}{4}(T^3+2T^2+T)} = \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T(T^2-1)}$$

$T \to \infty$, approach infinity

$\Rightarrow$ When $T \to \infty$, $\text{var}(b_2|x) = 0$ #

b.

$x_t = 0.5^t \Rightarrow \Sigma(x_t)^2 = \Sigma(0.5^t)^2 = \Sigma\, 0.25^t$

$$\text{var}(b_2|x) = \frac{\sigma^2}{\Sigma(x_t-\bar{x})^2} = \frac{\sigma^2}{\Sigma x_t^2 - T\bar{x}^2} = \frac{\sigma^2}{\Sigma x_t^2 - T(\frac{\Sigma x_t}{T})^2} = \frac{\sigma^2}{\Sigma\, 0.25^t - \frac{\Sigma\, 0.25^t}{T}}$$

公比 = 0.25

$$= \frac{\sigma^2}{\frac{0.25(1-0.25)}{1-0.25} - \frac{\frac{0.25(1-0.25)}{1-0.25}}{T}} = \frac{\sigma^2}{\frac{1}{3} - 0} \quad 3\sigma^2$$

$T \uparrow \infty$, approach $\frac{1}{3}$ $T \uparrow \infty$, approach 0

$\Rightarrow$ When $T \to \infty$, $\text{var}(b_2|x) = 3\sigma^2 \neq 0$ #

c.

For model a, the value of x increases over time, meaning the variation in x keeps growing. As the sample size approaches infinity, the variance of the estimator shrinks to zero, making it a consistent estimator.

For model b, the value of x decays exponentially. Eventually, x approaches zero and stops providing new variation. This implies that collecting more data in the future does not add useful information to the model. Therefore, the variance converges to $3\sigma^2$, making it an inconsistent estimator.

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \qquad \widehat{\operatorname{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

a. $\beta_2 = 0$
b. $\beta_1 + 2\beta_2 = 5$
c. $\beta_1 - \beta_2 + \beta_3 = 4$

$var(b_1) = 3$ $\quad cov(b_1, b_2) = -2$

$var(b_2) = 4$ $\quad cov(b_2, b_3) = 0$

$var(b_3) = 3$ $\quad cov(b_1, b_3) = 1$

5.6

a.

$H_0: \beta_2 = 0$, $H_1: \beta_2 \neq 0$

$t = \frac{b_2 - 0}{se(b_2)} \sim t_{60}$

$\Rightarrow t^* = \frac{3}{\sqrt{4}} = \frac{3}{2} = 1.5$

rejection region : $\{ |t| \geq 2.0003 \}$

Since t^* does not fall in the rejection region, we fail to reject the null hypothesis that $\beta_2 = 0$.

b.

$H_0: \beta_1 + 2\beta_2 = 5$, $H_1: \beta_1 + 2\beta_2 \neq 5$

$t = \frac{(b_1 + 2b_2) - 5}{se(b_1 + 2b_2)} \sim t_{60}$

$var(b_1 + 2b_2)$

$= var(b_1) + 4\,var(b_2) + 4\,cov(b_1, b_2)$

$= 3 + 4 \times 4 + 4 \times (-2) = 11$

$\Rightarrow t^* = \frac{(2 + 2 \times 3) - 5}{\sqrt{11}} = 0.90$

rejection region : $\{ |t| \geq 2.0003 \}$

Since t^* does not fall in the rejection region, we fail to reject the null hypothesis that $\beta_1 + 2\beta_2 = 5$

(-1) $\quad + 4\,cov(b_1, b_2)$

c. $H_0: \beta_1 - \beta_2 + \beta_3 = 4$, $H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$

$t = \frac{(b_1 - b_2 + b_3) - 4}{se(b_1 - b_2 + b_3)} \sim t_{60}$

$var(b_1 - b_2 + b_3)$

$= var(b_1) + var(b_2) + var(b_3) - 2\,cov(b_1, b_2) - 2\,cov(b_2, b_3)$

$+ 2\,cov(b_1, b_3)$

$= 3 + 4 + 3 + 4 - 0 + 2 = 16$

$\Rightarrow t^* = \frac{(2 - 3 - 1) - 4}{\sqrt{16}} = \frac{-6}{4} = -1.5$

rejection region : $\{ |t| \geq 2.0003 \}$

Since t^* does not fall in the rejection region, we fail to reject the null hypothesis that $\beta_1 - \beta_2 + \beta_3 = 4$.

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i$, $i = 1, 2, \ldots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x. Given that assumptions MR1–MR6 hold:

a. Show that $\hat{\beta}_{2,mean}$ is unbiased.

b. Derive an expression for $\text{var}\left(\hat{\beta}_{2,mean}|\mathbf{x}\right)$.

c. Write down an expression for $\text{var}\left(\hat{\beta}_{2,mean}\right)$.

d. Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2?

e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:

  i. $\text{var}(b_2|\mathbf{x})$ and $\text{var}\left(\hat{\beta}_{2,mean}|\mathbf{x}\right)$ where b_2 is the OLS estimator.

  ii. Estimates for $E\left[\left(s_x^2\right)^{-1}\right]$ and $E\left[4/\left(\bar{x}_2 - \bar{x}_1\right)^2\right]$ where s_x^2 is the sample standard deviation for x using N as a divisor.

f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?

g. Show that $E\left[\left(s_x^2\right)^{-1}\right] \xrightarrow{p} 0.12$ and $E\left[4/\left(\bar{x}_2 - \bar{x}_1\right)^2\right] \xrightarrow{p} 0.16$. [*Hint*: The variance of a uniform random variable defined on the interval (a, b) is $(b-a)^2/12$.]

h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

5.20

a.

prove that $E(\hat{\beta}_{2,mean}) = \beta_2$

$$\begin{cases} \bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1 & \text{(前半段資料的平均)} \\ \bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2 & \text{(後半段資料的平均)} \end{cases}$$

$$\hat{\beta}_{2,mean} \Rightarrow (\bar{y}_2 - \bar{y}_1) = \beta_2(\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)$$

$$\Rightarrow \frac{(\bar{y}_2 - \bar{y}_1)}{(\bar{x}_2 - \bar{x}_1)} = \beta_2 + \frac{(\bar{e}_2 - \bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)}$$

$$E(\hat{\beta}_{2,mean}|x) = \beta_2 + \frac{E[\bar{e}_2|x] - E[\bar{e}_1|x]}{(\bar{x}_2 - \bar{x}_1)}$$

MR2 : $E(e_i|x) = 0$

So $E[(\bar{e}_2 - \bar{e}_1)] = 0$

$\Rightarrow E(\hat{\beta}_{2,mean}) = \beta_2$, it is an unbiased estimator. #

b.

★ variance of sample mean $= \frac{\sigma^2}{n}$

$$\underset{\text{給定}x}{\text{var}(\hat{\beta}_{2,mean}|x)} = \text{var}\left(\beta_2 + \underbrace{\frac{(\bar{e}_2 - \bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)}}_{\text{constant}} \Big| x\right) = \text{var}(\beta_2) + \frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\text{var}(\bar{e}_2 - \bar{e}_1|x)$$

$\text{var}(\bar{e}_1) = \frac{\sigma^2}{N/2}$

$$= 0 + \frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\underbrace{[\text{var}(\bar{e}_2) + \text{var}(\bar{e}_1)]}_{\text{2組樣本獨立}} = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\left[\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N}\right] = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

c.

Law of total variance : $\text{Var}(Y) = E_x[\text{Var}(Y|X)] + \text{Var}_x(E[Y|X])$

$= \beta_2$ (proved in part a)

$\text{var}(\beta_2) = 0$

$$\text{var}\left(\hat{\beta}_2 + \frac{(\bar{e}_2 - \bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)}\right) = E_x\left[\text{Var}\left(\hat{\beta}_2 + \frac{(\bar{e}_2 - \bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)} \Big| x\right)\right] + \text{var}\left(E\left[\hat{\beta}_2 + \frac{(\bar{e}_2 - \bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)} \Big| x\right]\right)$$

$$= E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] = \frac{4\sigma^2}{N}E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right]$$ #

d.

Consistency : When we collect data, making the sample size N approach infinity, will the estimate stably converge to the true β_2 ?

对於一個不偏估計式，要達到一致性有一個必要條件，當 $N \to \infty$，它的 variance 必須逼近 0。

$$\lim_{N \to \infty} \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$ 整個分式會趨近於0

$(\bar{x}_2 - \bar{x}_1)^2 \neq 0$

=> 使 variance 收斂到 0，唯一的條件就是前一半和後一半的平均值不能相等。

g.

prove $E\left[(s_x^2)^{-1}\right] \to 0.12$

X是從區間(0,10)的均勻分配(uniform distribution)中抽出來的，均勻分配 $U(a,b)$ 的 var $= \frac{(b-a)^2}{12}$

$$\sigma_x^2 = \frac{(10-0)^2}{12} = \frac{25}{3}$$

取極限和倒數，$N \to \infty$ 時 $s_x^2 \to \sigma_x^2$

$$\lim_{N \to \infty} E\left[(s_x^2)^{-1}\right] = \frac{1}{\sigma_x^2} = \frac{1}{\frac{25}{3}} = \frac{3}{25} = 0.12 \quad \#$$

prove $E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \to 0.16$

- 前半段資料 $\bar{x}_1$：相當於從(0,5)的均勻分配(uniform distribution)中抽出來的，

  均勻分配 $U(a,b)$ 的 mean：$\frac{a+b}{2}$

  $$\bar{x}_1 \to \frac{0+5}{2} = 2.5$$

- 後半段資料 $\bar{x}_2$：相當於從(5,10)的均勻分配(uniform distribution)中抽出來的，

  均勻分配 $U(a,b)$ 的 mean：$\frac{a+b}{2}$

  $$\bar{x}_1 \to \frac{5+10}{2} = 7.5$$

$$\Rightarrow \lim_{N \to \infty} E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] = \frac{4}{(7.5-2.5)^2} = 0.16 \quad \#$$

5.20

e.

- The calculated variances and their corresponding components for different sample sizes are summarized in the table below:

Sample size (N)	$var(b_2\|x)$	$var(\hat{\beta}_{2,mean}\|x)$	$E[(s_x)^{-1}]$	$E[4/(\overline{x_2}-\overline{x_1})]^2$
100	1.2520	1.7112	0.1252	0.1711
500	0.2445	0.3327	0.1222	0.1663
1000	01218	0.1616	0.1218	0.1616
5000	0.024	0.0323	0.1209	0.1614

f.

- Comparing the values in part(e), the OLS variances are consistently smaller than the professor' s alternative variances across all sample sizes. This result illustrates the **Gauss-Markov Theorem**, which states that OLS is the **Best Linear Unbiased Estimator**, thus having the minimum variance.
- Furthermore, as the sample size increases, the professor' s variance approaches zero, indicating that this alternative estimator is still consistent. This consistency holds because we intentionally ordered the values of x before splitting the sample. Sorting ensures that the difference between upper and lower group means, $(\overline{x_2}-\overline{x_1})$, remains non-zero, keeping the denominator of the variance stable.

h.

- If we were to randomly assign the data into two groups without ordering the x values first, the expected values of $\overline{x_1}$ and $\overline{x_2}$ would both simply converge to the population mean of x.
- Consequently, as the sample size grows, the difference $(\overline{x_2}-\overline{x_1})$ in the denominator would converge to zero. Dividing by a value approaching zero would cause the estimator' s variance to explode toward infinity rather than shrinking to zero. Therefore, under random assignment, the professor' s estimator would **no longer be consistent**.